RTL Design Sherpa

RTL AMBA AXI4-Stream Module Reference — Rev 0.90

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Table of Contents

1 AXIS4 Clock-Gated Variants Guide

Location: `rtl/amba/axis4/*_cg.sv` **Status:** √ Production Ready

1.1 Overview

Both AXIS4 modules have clock-gated (`_cg`) variants that add power management through dynamic clock gating. Same architecture as AXI4/AXIL4 clock gating, optimized for streaming protocols.

1.1.1 Available Clock-Gated Modules

Module	Base Module	Description
axis_master_cg	axis_master	Clock-gated stream master
axis_slave_cg	axis_slave	Clock-gated stream slave

1.1.2 Key Features

- √ **Dynamic Clock Gating:** Automatic clock disable during idle
- √ **Configurable Idle Threshold:** Programmable idle count before gating
- √ **Full Functional Equivalence:** Identical to base modules
- √ **Status Monitoring:** Real-time gating status and clock count
- √ **Test Mode Support:** Bypass for scan testing
- √ **Zero Performance Impact:** Immediate ungating on activity

1.2 Additional Parameters (All _cg Modules)

Parameter	Type	Default	Description
CG_IDLE_COUNT_WIDTH	int	4	Width of idle counter (max idle = 2^N-1 cycles)

All other parameters identical to base module.

1.3 Additional Ports (All _cg Modules)

1.3.1 Clock Gating Configuration

Port	Direction	Width	Description
cfg_cg_enable	Input	1	Enable clock gating (0=disabled, 1=enabled)
cfg_cg_idle_count	Input	CG_IDLE_COUNT_WIDTH	Idle cycles before gating

1.3.2 Clock Gating Status

Port	Direction	Width	Description
cg_gating	Output	1	Clock currently gated (1=gated, 0=running)

Port	Direction	Width	Description
`cg_clk_count`	Output	32	Cumulative gated clock cycles

All other ports identical to base module.

1.4 Usage Example

```
axis_master_cg #(
    // Base parameters
    .SKID_DEPTH(4),
    .AXIS_DATA_WIDTH(64),
    .AXIS_ID_WIDTH(8),
    .AXIS_DEST_WIDTH(4),
    .AXIS_USER_WIDTH(1),

    // Clock gating
    .CG_IDLE_COUNT_WIDTH(4)  // Up to 15 idle cycles
) u_axis_master_cg (
    .aclk(stream_clk),
    .aresetn(stream_resetn),

    // Clock gating configuration
    .cfg_cg_enable(1'b1),          // Enable gating
    .cfg_cg_idle_count(4'd3),      // Gate after 3 idle cycles

    // AXIS signals (identical to base module)
    .fub_axis_tdata(src_tdata),
    // ... rest of AXIS signals ...

    // Clock gating status
    .cg_gating(stream_clk_gated),
    .cg_clk_count(stream_gated_cycles)
);
```

1.5 Clock Gating Behavior

Gating Conditions (All Must Be True): 1. Stream idle (no TVALID asserted) 2. Idle for ≥ `cfg_cg_idle_count` consecutive cycles 3. Gating enabled (`cfg_cg_enable = 1`)

Ungating Conditions (Any Triggers Immediate Ungating): 1. TVALID asserted (stream activity) 2. Configuration change

Ungating Latency: 0 cycles (immediate)

1.6 Configuration Guidelines

1.6.1 Idle Count Selection for Streaming

Aggressive Gating (Burst Packets):

```
.cfg_cg_idle_count(4'd1)   // Gate after 1 idle cycle
// Use for: Packet networks with gaps between packets
```

Moderate Gating (Mixed Traffic):

```
.cfg_cg_idle_count(4'd5)   // Gate after 5 idle cycles
// Use for: Video processing with blanking periods
```

Conservative Gating (Continuous Streams):

```
.cfg_cg_idle_count(4'd10)  // Gate after 10 idle cycles
// Use for: Low-latency continuous data flows
```

Disable Gating:

```
.cfg_cg_enable(1'b0)
// Use for: 100% utilization continuous streams
```

1.6.2 When to Use Clock Gating

✓ **Recommended For:** - Packet-based streaming (idle between packets) - Video processing (blanking periods between frames/lines) - Burst data transfers (gaps between bursts) - Power-constrained streaming applications

✗ **Not Recommended For:** - Continuous audio/video streams (100% utilization) - Real-time DSP pipelines (no idle periods) - Ultra-low-latency paths

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1.7 Power Savings Estimates

1.7.1 Typical Savings (Streaming Patterns)

Traffic Pattern	Duty Cycle	Idle Count	Power Savings
Packet network (1500B packets)	40%	1	30-35%

Traffic Pattern	Duty Cycle	Idle Count	Power Savings
Video (1080p with blanking)	70%	5	10-15%
Burst transfers	30%	1	35-40%
Continuous stream	100%	any	0%

Note: Streaming typically has longer continuous active periods vs register access, resulting in lower power savings than AXI4-Lite.

1.8 Complete Documentation

For full clock gating details, see: - **AXI4 Clock Gating Guide** - Complete reference - **AXIL4 Clock Gating Guide** - Additional examples

AXIS-specific notes: - Same architecture as AXI4/AXIL4 clock gating - Power savings vary based on stream duty cycle - Best for packet-based or frame-based streams

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1.9 Related Documentation

1.9.1 Base Modules

- **axis_master** - Base stream master
- **axis_slave** - Base stream slave

1.9.2 Architecture

- **AXI4 Clock Gating Guide** - Complete reference
- **AXIS4 Index** - AXIS4 module index

Last Updated: 2025-10-20

1.10 Navigation

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2 AXIS Master Interface (Clock-Gated)

Module: `axis_master_cg.sv` **Base Module:** axis_master **Location:** `rtl/amba/axis4/` **Status:** √ Production Ready

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2.1 Quick Reference

This is the **clock-gated variant** of axis_master.

For complete clock-gating documentation, usage examples, and configuration guidelines, see:

→ Clock-Gated Variants Guide

2.2 Summary

The `axis_master_cg` module adds power optimization to `axis_master` through activity-based clock gating:

- √ **Same Functionality:** 100% equivalent to base module
- √ **Power Savings:** 25-70% depending on traffic utilization
- √ **Configurable:** Idle threshold, gating domains, enable/disable
- √ **Zero Overhead When Disabled:** `ENABLE_CLOCK_GATING=0` → identical to base

2.3 Common Parameters

In addition to all axis_master parameters:

Parameter	Default	Description
`ENABLE_CLOCK_GATING`	1	Master enable (0=disable, identical to base)
`CG_IDLE_CYCLES`	8	Cycles before clock gating activates
`CG_GATE_*`	1	Domain-specific gating enables

2.4 Quick Usage

```
axis_master_cg #(
    // Base module parameters (see axis_master.md)
    .AXI_ID_WIDTH(8),
    .AXI_ADDR_WIDTH(32),
    .AXI_DATA_WIDTH(64),

    // Clock gating (see CG guide for details)
    .ENABLE_CLOCK_GATING(1),
    .CG_IDLE_CYCLES(8)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),
    // ... all other ports same as axis_master
);
```

2.5 Documentation

- **Base Module Functionality:** axis_master.md
- **Clock Gating Guide:** clock_gated_variants.md
- **Detailed CG Examples:**
  - axi4_master_rd_mon_cg.md (AXI4 monitor)
  - axil4_master_rd_mon_cg.md (AXIL4 monitor)
  - apb_slave_cg.md (APB interface)

2.6 Navigation

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3 axis_master

An AXI4-Stream master module that provides high-throughput streaming data transmission with configurable buffering and comprehensive support for all AXI4-Stream sideband signals including ID, DEST, and USER channels.

3.1 Overview

The `axis_master` module implements a complete AXI4-Stream master interface with integrated skid buffering for optimal streaming performance. It supports the full AXI4-Stream protocol with configurable data widths, optional sideband signals, and intelligent buffer management to maximize throughput in streaming data applications such as video processing, network packet handling, and DSP pipelines.

3.2 Module Declaration

```
module axis_master #(
    parameter int SKID_DEPTH         = 4,
    parameter int AXIS_DATA_WIDTH    = 32,
    parameter int AXIS_ID_WIDTH      = 8,
    parameter int AXIS_DEST_WIDTH    = 4,
    parameter int AXIS_USER_WIDTH    = 1,

    // Short and calculated params
    parameter int DW       = AXIS_DATA_WIDTH,
    parameter int IW       = AXIS_ID_WIDTH,
    parameter int DESTW    = AXIS_DEST_WIDTH,
    parameter int UW       = AXIS_USER_WIDTH,
    parameter int SW       = DW / 8,
    parameter int IW_WIDTH = (IW > 0) ? IW : 1,    // Minimum 1 bit
for zero-width signals
    parameter int DESTW_WIDTH = (DESTW > 0) ? DESTW : 1,
    parameter int UW_WIDTH = (UW > 0) ? UW : 1,
    parameter int TSize    = DW+SW+1+IW_WIDTH+DESTW_WIDTH+UW_WIDTH  //
Total packet size
) (
    // Global Clock and Reset
    input  logic                        aclk,
    input  logic                        aresetn,

    // Slave AXI4-Stream Interface (Input Side - FUB)
    input  logic [DW-1:0]               fub_axis_tdata,
    input  logic [SW-1:0]               fub_axis_tstrb,
    input  logic                        fub_axis_tlast,
    input  logic [IW_WIDTH-1:0]         fub_axis_tid,
    input  logic [DESTW_WIDTH-1:0]      fub_axis_tdest,
    input  logic [UW_WIDTH-1:0]         fub_axis_tuser,
    input  logic                        fub_axis_tvalid,
    output logic                        fub_axis_tready,

    // Master AXI4-Stream Interface (Output Side)
    output logic [DW-1:0]               m_axis_tdata,
    output logic [SW-1:0]               m_axis_tstrb,
```

```
    output logic                        m_axis_tlast,
    output logic [IW_WIDTH-1:0]         m_axis_tid,
    output logic [DESTW_WIDTH-1:0]      m_axis_tdest,
    output logic [UW_WIDTH-1:0]         m_axis_tuser,
    output logic                        m_axis_tvalid,
    input  logic                        m_axis_tready,

    // Status outputs for clock gating
    output logic                        busy
);
```

3.3 Parameters

Parameter	Type	Default	Description
SKID_DEPTH	int	4	Skid buffer depth (2^N entries)
AXIS_DATA_WIDTH	int	32	AXI4-Stream data bus width (bytes)
AXIS_ID_WIDTH	int	8	Stream ID width (0 to disable)
AXIS_DEST_WIDTH	int	4	Destination width (0 to disable)
AXIS_USER_WIDTH	int	1	User signal width (0 to disable)

3.4 Ports

3.4.1 Clock and Reset

Port	Width	Direction	Description
aclk	1	Input	AXI4-Stream clock
aresetn	1	Input	AXI4-Stream active-low reset

3.4.2 Slave AXI4-Stream Interface (Input Side - FUB)

Port	Width	Direction	Description
fub_axis_tdata	AXIS_DATA_WIDTH	Input	Stream data
fub_axis_tstrb	AXIS_DATA_WIDTH/8	Input	Data byte strobes
fub_axis_tlast	1	Input	Last transfer in packet
fub_axis_tid	AXIS_ID_WIDTH	Input	Stream identifier
fub_axis_tdest	AXIS_DEST_WIDTH	Input	Destination routing
fub_axis_tuser	AXIS_USER_WIDTH	Input	User-defined sideband
fub_axis_tvalid	1	Input	Transfer valid
fub_axis_tready	1	Output	Transfer ready

3.4.3 Master AXI4-Stream Interface (Output Side)

Port	Width	Direction	Description
m_axis_tdata	AXIS_DATA_WIDTH	Output	Stream data
m_axis_tstrb	AXIS_DATA_WIDTH/8	Output	Data byte strobes
m_axis_tlast	1	Output	Last transfer in packet
m_axis_tid	AXIS_ID_WIDTH	Output	Stream identifier
m_axis_tdest	AXIS_DEST_WIDTH	Output	Destination routing
m_axis_tuser	AXIS_USER_WIDTH	Output	User-defined sideband
m_axis_tvalid	1	Output	Transfer valid
m_axis_tready	1	Input	Transfer ready

3.4.4 Status Interface

Port	Width	Direction	Description
busy	1	Output	Module activity indicator for clock gating

3.5 Architecture

3.5.1 Skid Buffer Design

The module employs a single GAXI skid buffer to manage streaming data flow:

1. **Input Buffering**: Incoming stream data buffered for flow control
2. **Packet Packing**: All stream signals packed into single buffer entry
3. **Flexible Unpacking**: Conditional unpacking based on enabled sideband signals

3.5.2 Data Flow

```
FUB AXIS → Skid Buffer → Master AXIS → Downstream
     ↑           ↓                ↓
    Ready    Flow Control    Ready/Valid
```

3.5.3 Key Features

- **AXI4-Stream Compliance**: Full protocol support including all optional signals
- **Flexible Configuration**: Zero-width support for unused sideband signals
- **Flow Control**: Skid buffer prevents pipeline stalls
- **Clock Gating Support**: Busy signal for power optimization
- **Packet Integrity**: TLAST preservation through buffering
- **Multi-Stream Support**: TID and TDEST routing capabilities

3.6 Signal Description

3.6.1 Core Signals

Signal	Width	Description
TDATA	8-512 bits	Primary data payload

Signal	Width	Description
TSTRB	TDATA/8 bits	Byte lane strobes (1=valid byte)
TVALID	1 bit	Transfer valid indicator
TREADY	1 bit	Transfer ready (backpressure)
TLAST	1 bit	Last transfer in packet/frame

3.6.2 Optional Sideband Signals

Signal	Width	Description
TID	0-16 bits	Stream identifier for multiplexing
TDEST	0-16 bits	Destination routing information
TUSER	0-16 bits	User-defined control/status

3.7 Functionality

3.7.1 Buffer Management

The skid buffer provides: 1. **Decoupling**: Separates upstream and downstream timing 2. **Flow Control**: Prevents data loss during backpressure 3. **Pipeline Optimization**: Eliminates ready-path combinatorial logic

3.7.2 Conditional Signal Handling

The module uses generate blocks to handle various combinations of enabled/disabled sideband signals:

```
// Example: Full signals enabled
if (IW > 0 && DESTW > 0 && UW > 0) begin
    // All sideband signals active
end else if (IW > 0 && DESTW == 0 && UW == 0) begin
    // Only TID active
end
// ... additional combinations
```

3.7.3 Busy Signal Generation

Activity detection for clock gating:

```
busy = (buffer_count > 0) || fub_axis_tvalid;
```

3.8 Timing Characteristics

3.8.1 Buffer Performance

Characteristic	Value	Description
Buffer Depth	16 entries (default)	2^SKID_DEPTH entries
Buffering Latency	1 clock cycle	Input to output delay
Flow Control Latency	1 clock cycle	Ready propagation

3.8.2 Throughput Metrics

Metric	Value	Conditions
Maximum Frequency	400-800 MHz	Technology dependent
Peak Throughput	3.2-25.6 GB/s	32-bit at max frequency
Sustained Throughput	95-99% of peak	With proper buffering
Pipeline Efficiency	>95%	Continuous data flow

3.9 Usage Examples

3.9.1 Basic Video Stream Processing

```
axis_master #(
    .SKID_DEPTH(4),           // 16-entry buffer
    .AXIS_DATA_WIDTH(64),     // 8 bytes per transfer
    .AXIS_ID_WIDTH(4),        // Support 16 streams
    .AXIS_DEST_WIDTH(4),      // Support 16 destinations
    .AXIS_USER_WIDTH(8)       // 8-bit control signals
) u_video_stream (
    .aclk           (video_clk),
    .aresetn        (video_resetn),
```

```
    // Input from video source
    .fub_axis_tdata    (pixel_data),
    .fub_axis_tstrb    (pixel_strb),
    .fub_axis_tlast    (line_end),
    .fub_axis_tid      (stream_id),
    .fub_axis_tdest    (display_id),
    .fub_axis_tuser    (pixel_ctrl),
    .fub_axis_tvalid   (pixel_valid),
    .fub_axis_tready   (pixel_ready),

    // Output to video pipeline
    .m_axis_tdata      (pipe_tdata),
    .m_axis_tstrb      (pipe_tstrb),
    .m_axis_tlast      (pipe_tlast),
    .m_axis_tid        (pipe_tid),
    .m_axis_tdest      (pipe_tdest),
    .m_axis_tuser      (pipe_tuser),
    .m_axis_tvalid     (pipe_tvalid),
    .m_axis_tready     (pipe_tready),

    .busy              (video_busy)
);
```

3.9.2 Network Packet Processing

```
// High-performance packet processing
axis_master #(
    .SKID_DEPTH(5),           // 32-entry buffer for latency tolerance
    .AXIS_DATA_WIDTH(512),    // 64 bytes per beat (512-bit)
    .AXIS_ID_WIDTH(8),        // 256 flow IDs
    .AXIS_DEST_WIDTH(6),      // 64 output ports
    .AXIS_USER_WIDTH(16)      // Packet metadata
) u_packet_master (
    .aclk            (net_clk),
    .aresetn         (net_resetn),

    // Input from packet classifier
    .fub_axis_tdata    (pkt_data),
    .fub_axis_tstrb    (pkt_keep),
    .fub_axis_tlast    (pkt_eop),
    .fub_axis_tid      (flow_id),
    .fub_axis_tdest    (output_port),
    .fub_axis_tuser    ({pkt_len, pkt_type}),
    .fub_axis_tvalid   (pkt_valid),
    .fub_axis_tready   (pkt_ready),

    // Output to switching fabric
```

```
    .m_axis_*(switch_axis_*),

    .busy               (pkt_proc_busy)
);
```

3.9.3 DSP Data Pipeline

```
// DSP processing chain with minimal sideband
axis_master #(
    .SKID_DEPTH(3),           // 8-entry buffer
    .AXIS_DATA_WIDTH(128),    // 4 x 32-bit samples
    .AXIS_ID_WIDTH(0),        // No stream ID needed
    .AXIS_DEST_WIDTH(0),      // No destination routing
    .AXIS_USER_WIDTH(4)       // Sample metadata
) u_dsp_stream (
    .aclk            (dsp_clk),
    .aresetn         (dsp_resetn),

    // Input from ADC interface
    .fub_axis_tdata    (adc_samples),
    .fub_axis_tstrb    (adc_valid_bytes),
    .fub_axis_tlast    (frame_end),
    .fub_axis_tid      (1'b0),          // Unused
    .fub_axis_tdest    (1'b0),          // Unused
    .fub_axis_tuser    (sample_metadata),
    .fub_axis_tvalid   (adc_valid),
    .fub_axis_tready   (adc_ready),

    // Output to DSP processing
    .m_axis_tdata      (proc_samples),
    .m_axis_tstrb      (proc_strb),
    .m_axis_tlast      (proc_frame_end),
    .m_axis_tid        (),              // Unconnected
    .m_axis_tdest      (),              // Unconnected
    .m_axis_tuser      (proc_metadata),
    .m_axis_tvalid     (proc_valid),
    .m_axis_tready     (proc_ready),

    .busy              (dsp_busy)
);
```

3.9.4 Multi-Stream Router Integration

```
module stream_router (
    input logic axi_clk,
    input logic axi_resetn,

    // Multiple input streams
```

```
    axi4s_if.slave  input_streams [4],
    // Single output stream
    axi4s_if.master output_stream
);

    // Stream masters for each input with buffering
    genvar i;
    generate
        for (i = 0; i < 4; i++) begin : gen_stream_masters
            axis_master #(
                .SKID_DEPTH(4),
                .AXIS_DATA_WIDTH(64),
                .AXIS_ID_WIDTH(4),
                .AXIS_DEST_WIDTH(4),
                .AXIS_USER_WIDTH(1)
            ) u_stream_master (
                .aclk(axi_clk),
                .aresetn(axi_resetn),

                // Connect to input stream
                .fub_axis_*(input_streams[i].*),

                // Connect to arbiter
                .m_axis_*(arb_inputs[i].*),

                .busy(stream_busy[i])
            );
        end
    endgenerate

    // Stream arbiter for output selection
    axis_arbiter #(
        .NUM_INPUTS(4),
        .DATA_WIDTH(64)
    ) u_arbiter (
        .aclk(axi_clk),
        .aresetn(axi_resetn),
        .s_axis(arb_inputs),
        .m_axis(output_stream),
        .active_mask(stream_enable)
    );

endmodule
```

3.10 Advanced Integration Patterns

3.10.1 Clock Domain Crossing

```
// Cross clock domains with async FIFO
module axis_cdc_system (
    // Source domain
    input  logic        src_clk,
    input  logic        src_resetn,
    axi4s_if.slave      src_axis,

    // Destination domain
    input  logic        dst_clk,
    input  logic        dst_resetn,
    axi4s_if.master     dst_axis
);

    // Source domain buffering
    axis_master #(
        .SKID_DEPTH(3),
        .AXIS_DATA_WIDTH(32)
    ) u_src_master (
        .aclk(src_clk),
        .aresetn(src_resetn),
        .fub_axis_*(src_axis.*),
        .m_axis_*(cdc_src_axis.*),
        .busy(src_busy)
    );

    // Async clock domain crossing
    gaxi_fifo_async #(
        .DEPTH(6),  // 64 entries
        .DATA_WIDTH(32+4+1+4+4+1)
    ) u_cdc_fifo (
        .wr_clk(src_clk),
        .wr_resetn(src_resetn),
        .wr_axis(cdc_src_axis),

        .rd_clk(dst_clk),
        .rd_resetn(dst_resetn),
        .rd_axis(dst_axis)
    );

endmodule
```

3.10.2 Stream Processing Pipeline

```
// Multi-stage processing pipeline
module stream_pipeline (
    input logic clk, resetn,
    axi4s_if.slave  input_stream,
    axi4s_if.master output_stream
);

    // Pipeline stage interfaces
    axi4s_if stage1_out();
    axi4s_if stage2_out();
    axi4s_if stage3_out();

    // Stage 1: Input buffering
    axis_master #(.SKID_DEPTH(3), .AXIS_DATA_WIDTH(64))
    u_stage1 (
        .aclk(clk), .aresetn(resetn),
        .fub_axis_*(input_stream.*),
        .m_axis_*(stage1_out.*),
        .busy(stage1_busy)
    );

    // Stage 2: Processing with buffering
    stream_processor u_processor (
        .clk(clk), .resetn(resetn),
        .s_axis(stage1_out), .m_axis(stage2_out)
    );

    axis_master #(.SKID_DEPTH(4), .AXIS_DATA_WIDTH(64))
    u_stage2 (
        .aclk(clk), .aresetn(resetn),
        .fub_axis_*(stage2_out.*),
        .m_axis_*(stage3_out.*),
        .busy(stage2_busy)
    );

    // Stage 3: Output conditioning
    axis_master #(.SKID_DEPTH(2), .AXIS_DATA_WIDTH(64))
    u_stage3 (
        .aclk(clk), .aresetn(resetn),
        .fub_axis_*(stage3_out.*),
        .m_axis_*(output_stream.*),
        .busy(stage3_busy)
    );

    assign pipeline_busy = stage1_busy | stage2_busy | stage3_busy;
```

```
endmodule
```

3.11 Performance Optimization

3.11.1 Buffer Depth Selection

Choose buffer depths based on application characteristics:

Application Type	Recommended DEPTH	Buffer Size	Rationale
Low Latency DSP	2-3 (4-8 entries)	Minimal	Reduce processing delay
Video Streaming	4-5 (16-32 entries)	Medium	Handle line buffers
Network Packets	5-6 (32-64 entries)	Large	Variable packet sizes
High Throughput	6-7 (64-128 entries)	Very Large	Maximum bandwidth

3.11.2 Data Width Optimization

```
// Optimize data width for application
localparam VIDEO_PIXELS_PER_CYCLE = 4;
localparam PIXEL_BITS = 24;  // RGB888
localparam VIDEO_DATA_WIDTH = VIDEO_PIXELS_PER_CYCLE * PIXEL_BITS;

axis_master #(
    .AXIS_DATA_WIDTH(VIDEO_DATA_WIDTH),  // 96 bits
    .SKID_DEPTH(4),
    .AXIS_USER_WIDTH(8)  // Control signals
) u_video_master (...);
```

3.11.3 Sideband Signal Optimization

```
// Minimize unused sideband signals for area/power
axis_master #(
    .AXIS_DATA_WIDTH(128),
    .AXIS_ID_WIDTH(0),     // Disable TID
    .AXIS_DEST_WIDTH(0),   // Disable TDEST
    .AXIS_USER_WIDTH(4),   // Minimal TUSER
    .SKID_DEPTH(3)
) u_optimized_master (...);
```

3.12 Clock Gating Integration

```
// Clock gated version for power optimization
axis_master_cg #(
    .SKID_DEPTH(4),
    .AXIS_DATA_WIDTH(64)
) u_cg_master (
    .aclk            (axi_clk),
    .aresetn         (axi_resetn),

    // Standard AXI4-Stream interfaces
    .fub_axis_*(fub_axis_*),
    .m_axis_*(m_axis_*),

    // Clock gating control
    .cg_enable       (stream_enable),
    .cg_test_enable  (scan_mode),
    .busy            (stream_busy)
);

// System-level power management
always_ff @(posedge axi_clk) begin
    stream_power_down <= !stream_busy && (idle_time >
POWER_DOWN_DELAY);
end
```

3.13 Synthesis Considerations

3.13.1 Area Optimization

- Use minimum required data widths
- Disable unused sideband signals (set width to 0)
- Optimize buffer depths for application requirements
- Share buffers across multiple streams when possible

3.13.2 Timing Optimization

- Register all interface outputs for timing closure
- Use appropriate buffer depths to break critical paths
- Consider pipeline stages for very high-frequency designs

3.13.3 Power Optimization

- Use clock gating variant (`axis_master_cg`) when available
- Implement activity-based power scaling

- Size buffers appropriately to minimize switching

3.14 Verification Notes

3.14.1 Protocol Compliance

- Verify AXI4-Stream handshaking (VALID/READY)
- Check TLAST alignment with packet boundaries
- Validate sideband signal preservation

3.14.2 Buffer Verification

- Test buffer overflow/underflow protection
- Verify data integrity through buffering
- Check flow control under backpressure

3.14.3 Performance Verification

- Measure sustained throughput under various loads
- Verify buffer utilization efficiency
- Check latency characteristics

3.15 Related Modules

- **axis_master_cg**: Clock-gated version for power optimization
- **axis_slave**: Complementary AXI4-Stream slave implementation
- **gaxi_skid_buffer**: Underlying buffer infrastructure
- **gaxi_fifo_async**: Asynchronous FIFO for clock domain crossing
- **axis_interconnect**: AXI4-Stream interconnect for routing

The `axis_master` module provides a complete, high-performance solution for AXI4-Stream master functionality with advanced buffering, flexible signal configuration, and comprehensive system integration capabilities.

4 AXIS Slave Interface (Clock-Gated)

Module: `axis_slave_cg.sv` **Base Module:** axis_slave **Location:** `rtl/amba/axis4/` **Status:** ✓ Production Ready

---

4.1 Quick Reference

This is the **clock-gated variant** of axis_slave.

For complete clock-gating documentation, usage examples, and configuration guidelines, see:

→ Clock-Gated Variants Guide

4.2 Summary

The `axis_slave_cg` module adds power optimization to `axis_slave` through activity-based clock gating:

- √ **Same Functionality:** 100% equivalent to base module
- √ **Power Savings:** 25-70% depending on traffic utilization
- √ **Configurable:** Idle threshold, gating domains, enable/disable
- √ **Zero Overhead When Disabled:** `ENABLE_CLOCK_GATING=0` → identical to base

4.3 Common Parameters

In addition to all axis_slave parameters:

Parameter	Default	Description
`ENABLE_CLOCK_GATING`	1	Master enable (0=disable, identical to base)
`CG_IDLE_CYCLES`	8	Cycles before clock gating activates
`CG_GATE_*`	1	Domain-specific gating enables

4.4 Quick Usage

```
axis_slave_cg #(
    // Base module parameters (see axis_slave.md)
    .AXI_ID_WIDTH(8),
    .AXI_ADDR_WIDTH(32),
```

```
    .AXI_DATA_WIDTH(64),

    // Clock gating (see CG guide for details)
    .ENABLE_CLOCK_GATING(1),
    .CG_IDLE_CYCLES(8)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),
    // ... all other ports same as axis_slave
);
```

4.5 Documentation

- **Base Module Functionality:** axis_slave.md
- **Clock Gating Guide:** clock_gated_variants.md
- **Detailed CG Examples:**
  - axi4_master_rd_mon_cg.md (AXI4 monitor)
  - axil4_master_rd_mon_cg.md (AXIL4 monitor)
  - apb_slave_cg.md (APB interface)

4.6 Navigation

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5 axis_slave

An AXI4-Stream slave module that provides high-throughput streaming data reception with configurable buffering and comprehensive support for all AXI4-Stream sideband signals including ID, DEST, and USER channels.

5.1 Overview

The `axis_slave` module implements a complete AXI4-Stream slave interface with integrated skid buffering for optimal streaming performance. It supports the full AXI4-Stream protocol with configurable data widths, optional sideband signals, and intelligent buffer management for streaming data applications.

5.2 Module Declaration

```
module axis_slave #(
    parameter int SKID_DEPTH         = 4,
    parameter int AXIS_DATA_WIDTH    = 32,
    parameter int AXIS_ID_WIDTH      = 8,
    parameter int AXIS_DEST_WIDTH    = 4,
    parameter int AXIS_USER_WIDTH    = 1,

    // Short and calculated params
    parameter int DW       = AXIS_DATA_WIDTH,
    parameter int IW       = AXIS_ID_WIDTH,
    parameter int DESTW    = AXIS_DEST_WIDTH,
    parameter int UW       = AXIS_USER_WIDTH,
    parameter int SW       = DW / 8,
    parameter int IW_WIDTH = (IW > 0) ? IW : 1,
    parameter int DESTW_WIDTH = (DESTW > 0) ? DESTW : 1,
    parameter int UW_WIDTH = (UW > 0) ? UW : 1,
    parameter int TSize    = DW+SW+1+IW_WIDTH+DESTW_WIDTH+UW_WIDTH
) (
    // Global Clock and Reset
    input  logic                        aclk,
    input  logic                        aresetn,

    // Slave AXI4-Stream Interface (Input Side from interconnect)
    input  logic [DW-1:0]               s_axis_tdata,
    input  logic [SW-1:0]               s_axis_tstrb,
    input  logic                        s_axis_tlast,
    input  logic [IW_WIDTH-1:0]         s_axis_tid,
    input  logic [DESTW_WIDTH-1:0]      s_axis_tdest,
    input  logic [UW_WIDTH-1:0]         s_axis_tuser,
    input  logic                        s_axis_tvalid,
    output logic                        s_axis_tready,

    // Master AXI4-Stream Interface (Output Side to backend)
    output logic [DW-1:0]               fub_axis_tdata,
    output logic [SW-1:0]               fub_axis_tstrb,
    output logic                        fub_axis_tlast,
    output logic [IW_WIDTH-1:0]         fub_axis_tid,
    output logic [DESTW_WIDTH-1:0]      fub_axis_tdest,
    output logic [UW_WIDTH-1:0]         fub_axis_tuser,
    output logic                        fub_axis_tvalid,
    input  logic                        fub_axis_tready,

    // Status outputs for clock gating
    output logic                        busy
);
```

5.3 Parameters

Parameter	Type	Default	Description
SKID_DEPTH	int	4	Skid buffer depth (2^N entries)
AXIS_DATA_WIDTH	int	32	AXI4-Stream data bus width (bytes)
AXIS_ID_WIDTH	int	8	Stream ID width (0 to disable)
AXIS_DEST_WIDTH	int	4	Destination width (0 to disable)
AXIS_USER_WIDTH	int	1	User signal width (0 to disable)

5.4 Ports

5.4.1 Slave AXI4-Stream Interface (Input from Interconnect)

Port	Width	Direction	Description
s_axis_tdata	DW	Input	Stream data
s_axis_tstrb	SW	Input	Data strobes (byte-valid indicators)
s_axis_tlast	1	Input	Last transfer in packet
s_axis_tid	IW	Input	Stream ID (routing/reordering)
s_axis_tdest	DESTW	Input	Destination routing
s_axis_tuser	UW	Input	User-defined sideband

Port	Width	Direction	Description
s_axis_tvalid	1	Input	Data valid
s_axis_tready	1	Output	Ready to accept data

5.4.2 Backend Interface (Output to Processing Logic)

Port	Width	Direction	Description
fub_axis_tdata	DW	Output	Stream data (to backend)
fub_axis_tstrb	SW	Output	Data strobes
fub_axis_tlast	1	Output	Last transfer in packet
fub_axis_tid	IW	Output	Stream ID
fub_axis_tdest	DESTW	Output	Destination routing
fub_axis_tuser	UW	Output	User-defined sideband
fub_axis_tvalid	1	Output	Data valid (to backend)
fub_axis_tready	1	Input	Backend ready

5.4.3 Status

Port	Width	Direction	Description
busy	1	Output	Interface active (for clock gating)

5.5 Features

- **Full AXI4-Stream Protocol:** Support for all optional sideband signals
- **Elastic Buffering:** Skid buffer decouples interconnect and backend timing
- **Configurable Width:** Optional ID, DEST, USER signals (set width to 0 to disable)
- **Activity Monitoring:** Busy signal for power management

5.6 Usage Example

```
axis_slave #(
    .SKID_DEPTH(4),           // 16 entries
    .AXIS_DATA_WIDTH(64),
    .AXIS_ID_WIDTH(8),
    .AXIS_DEST_WIDTH(4),
    .AXIS_USER_WIDTH(1)
) u_axis_slave (
    .aclk(stream_clk),
    .aresetn(stream_resetn),

    // Slave interface (from network/interconnect)
    .s_axis_tdata(net_tdata),
    .s_axis_tstrb(net_tstrb),
    .s_axis_tlast(net_tlast),
    .s_axis_tid(net_tid),
    .s_axis_tdest(net_tdest),
    .s_axis_tuser(net_tuser),
    .s_axis_tvalid(net_tvalid),
    .s_axis_tready(net_tready),

    // Backend interface (to processing logic)
    .fub_axis_tdata(proc_tdata),
    .fub_axis_tstrb(proc_tstrb),
    .fub_axis_tlast(proc_tlast),
    .fub_axis_tid(proc_tid),
    .fub_axis_tdest(proc_tdest),
    .fub_axis_tuser(proc_tuser),
    .fub_axis_tvalid(proc_tvalid),
    .fub_axis_tready(proc_tready),

    .busy(slave_busy)
);
```

5.7 Design Notes

- **Signal Flow:** Interconnect → Skid Buffer → Backend
- **Use Case:** Stream receivers, packet processors, video decoders
- **Buffering:** Allows backend to consume at its own pace without stalling interconnect

5.8 Related Modules

- **axis_master** - Stream master counterpart
- **axis_slave_cg** - Clock-gated version

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5.9 Navigation

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6 AXIS4 (AXI4-Stream) Modules

Location: `rtl/amba/axis4/` **Test Location:** `val/amba/` **Status:** √ Production Ready

6.1 Overview

The AXIS4 subsystem provides a complete implementation of the ARM AMBA AXI4-Stream protocol—a high-performance streaming interface optimized for unidirectional data flows such as video processing, network packet handling, and DSP pipelines.

6.1.1 Key Features

- √ **AXI4-Stream Protocol:** Streaming-optimized subset (no address channels)
- √ **High Throughput:** Up to 25.6 GB/s (512-bit @ 400 MHz)
- √ **Flexible Sideband Signals:** TID, TDEST, TUSER for routing/control
- √ **Packet Framing:** TLAST for packet/frame boundaries
- √ **Elastic Buffering:** Integrated skid buffers for timing closure
- √ **Clock Gating Variants:** Power-optimized versions
- √ **Typical Use:** Video streams, network packets, DSP pipelines

6.2 Module Categories

6.2.1 Core Master/Slave Modules

Module	Description	Documentation	Status
axis_master	AXIS master with buffered output	axis_master.md	√ Documented
axis_slave	AXIS slave with	axis_slave.md	√

Module	Description	Documentation	Status
	buffered input		Documented

6.2.2 Clock-Gated Variants

All clock-gated variants documented in: axis_clock_gating_guide.md

Module	Base Module	Status
axis_master_cg	axis_master	√ Documented
axis_slave_cg	axis_slave	√ Documented

6.3 AXI4-Stream Protocol Overview

6.3.1 Streaming vs Memory-Mapped

Feature	AXI4	AXI4-Stream
Address Channels	AW/AR (5 channels)	None (streaming)
Data Flow	Bidirectional	Unidirectional
Transaction Type	Memory-mapped	Data streaming
Backpressure	Via ready signals	Via TREADY
Packet Framing	Not applicable	TLAST signal
Routing	Address-based	TID/TDEST-based
Typical Use	Register/memory access	Video/network/DSP

6.3.2 Signal Architecture

AXI4-Stream uses a single channel for streaming data:

Core Signals: - **TDATA** - Streaming data payload (8-512 bits) - **TSTRB** - Byte lane strobes (1 bit per byte) - **TVALID** - Data valid indicator - **TREADY** - Ready for data (backpressure) - **TLAST** - Last transfer in packet/frame

Optional Sideband Signals: - **TID** - Stream identifier (multiplexing) - **TDEST** - Destination routing - **TUSER** - User-defined control/status

6.3.3 Key Signals

Data Transfer: - `TDATA[N:0]` - Primary streaming data - `TSTRB[N/8:0]` - Byte valid indicators (1=valid byte) - `TVALID` - Transfer valid - `TREADY` - Transfer ready

Packet Control: - `TLAST` - End of packet/frame marker

Routing/Control: - `TID[N:0]` - Stream ID for multiplexing (optional) - `TDEST[N:0]` - Destination routing information (optional) - `TUSER[N:0]` - User-defined metadata (optional)

6.4 Quick Start

6.4.1 Basic Stream Master

```
axis_master #(
    .SKID_DEPTH(4),           // 16 entries
    .AXIS_DATA_WIDTH(64),     // 8 bytes per transfer
    .AXIS_ID_WIDTH(4),        // 16 streams
    .AXIS_DEST_WIDTH(4),      // 16 destinations
    .AXIS_USER_WIDTH(8)       // 8-bit control
) u_stream_master (
    .aclk               (stream_clk),
    .aresetn            (stream_resetn),

    // Frontend interface (from source)
    .fub_axis_tdata     (src_tdata),
    .fub_axis_tstrb     (src_tstrb),
    .fub_axis_tlast     (src_tlast),
    .fub_axis_tid       (src_tid),
    .fub_axis_tdest     (src_tdest),
    .fub_axis_tuser     (src_tuser),
    .fub_axis_tvalid    (src_tvalid),
    .fub_axis_tready    (src_tready),

    // Master interface (to interconnect)
    .m_axis_tdata       (net_tdata),
    // ... rest of master signals

    .busy               (master_busy)
);
```

6.4.2 Basic Stream Slave

```
axis_slave #(
    .SKID_DEPTH(4),           // 16 entries
    .AXIS_DATA_WIDTH(64),
    .AXIS_ID_WIDTH(8),
    .AXIS_DEST_WIDTH(4),
    .AXIS_USER_WIDTH(1)
) u_stream_slave (
    .aclk               (stream_clk),
```

```
    .aresetn            (stream_resetn),

    // Slave interface (from interconnect)
    .s_axis_tdata       (net_tdata),
    .s_axis_tstrb       (net_tstrb),
    .s_axis_tlast       (net_tlast),
    .s_axis_tid         (net_tid),
    .s_axis_tdest       (net_tdest),
    .s_axis_tuser       (net_tuser),
    .s_axis_tvalid      (net_tvalid),
    .s_axis_tready      (net_tready),

    // Backend interface (to processor)
    .fub_axis_tdata     (proc_tdata),
    // ... rest of backend signals

    .busy               (slave_busy)
);
```

6.4.3 Video Processing Pipeline

```
// Video frame buffer with packet boundaries
axis_master #(
    .SKID_DEPTH(5),           // 32-entry buffer
    .AXIS_DATA_WIDTH(96),     // 4 pixels x 24-bit RGB
    .AXIS_ID_WIDTH(0),        // No stream ID
    .AXIS_DEST_WIDTH(4),      // 16 display outputs
    .AXIS_USER_WIDTH(16)      // Frame/line metadata
) u_video_master (
    .aclk               (pixel_clk),
    .aresetn            (video_resetn),

    // From frame buffer
    .fub_axis_tdata     (pixel_data),      // RGB pixel data
    .fub_axis_tstrb     (pixel_strb),      // Valid bytes
    .fub_axis_tlast     (line_end),        // End of video line
    .fub_axis_tid       (1'b0),            // Unused
    .fub_axis_tdest     (display_id),      // Target display
    .fub_axis_tuser     ({frame_num, line_num}),
    .fub_axis_tvalid    (pixel_valid),
    .fub_axis_tready    (pixel_ready),

    // To video processing chain
    .m_axis_tdata       (proc_pixel_data),
    .m_axis_tstrb       (proc_pixel_strb),
    .m_axis_tlast       (proc_line_end),
    .m_axis_tid         (),
    .m_axis_tdest       (proc_display_id),
```

```
    .m_axis_tuser       (proc_metadata),
    .m_axis_tvalid      (proc_valid),
    .m_axis_tready      (proc_ready),

    .busy               (video_busy)
);
```

6.4.4 Network Packet Processing

```
// High-bandwidth packet processing
axis_slave #(
    .SKID_DEPTH(6),           // 64-entry buffer for latency
    .AXIS_DATA_WIDTH(512),    // 64 bytes per beat
    .AXIS_ID_WIDTH(8),        // 256 flow IDs
    .AXIS_DEST_WIDTH(6),      // 64 output ports
    .AXIS_USER_WIDTH(16)      // Packet metadata
) u_packet_slave (
    .aclk               (net_clk),
    .aresetn            (net_resetn),

    // From network interface
    .s_axis_tdata       (rx_pkt_data),
    .s_axis_tstrb       (rx_pkt_keep),     // Byte valid
    .s_axis_tlast       (rx_pkt_eop),      // End of packet
    .s_axis_tid         (rx_flow_id),
    .s_axis_tdest       (rx_output_port),
    .s_axis_tuser       ({pkt_len, pkt_type}),
    .s_axis_tvalid      (rx_pkt_valid),
    .s_axis_tready      (rx_pkt_ready),

    // To packet processor
    .fub_axis_tdata     (proc_pkt_data),
    .fub_axis_tstrb     (proc_pkt_keep),
    .fub_axis_tlast     (proc_pkt_eop),
    .fub_axis_tid       (proc_flow_id),
    .fub_axis_tdest     (proc_port),
    .fub_axis_tuser     (proc_metadata),
    .fub_axis_tvalid    (proc_pkt_valid),
    .fub_axis_tready    (proc_pkt_ready),

    .busy               (packet_busy)
);
```

6.5 Testing

All AXIS4 modules are verified using CocoTB-based testbenches:

```
# Run all AXIS4 tests
pytest val/amba/test_axis*.py -v

# Run specific module tests
pytest val/amba/test_axis_master.py -v
pytest val/amba/test_axis_slave.py -v

# Run clock-gated variant tests
pytest val/amba/test_axis_master_cg.py -v
pytest val/amba/test_axis_slave_cg.py -v

# Run with waveforms
pytest val/amba/test_axis_master.py --vcd=waves.vcd -v
```

6.6 Design Patterns

6.6.1 Pattern 1: Elastic Buffering

All core modules use `gaxi_skid_buffer`: - Decouples source and sink timing - Configurable depth per module - 1-cycle latency overhead - Full backpressure handling

6.6.2 Pattern 2: Packet Framing

TLAST signal marks packet boundaries:

```
// Video: TLAST marks end of line
tlast = (pixel_count == PIXELS_PER_LINE - 1);

// Network: TLAST marks end of packet
tlast = (byte_count == packet_length - 1);

// DSP: TLAST marks end of frame
tlast = (sample_count == FRAME_SIZE - 1);
```

6.6.3 Pattern 3: Stream Multiplexing

TID enables multiple logical streams:

```
// Four video streams on single physical interface
.fub_axis_tid(camera_id),  // 0-3 for 4 cameras
.fub_axis_tdest(display_id) // Route to specific display
```

6.6.4 Pattern 4: Clock Gating

Clock-gated variants (*_cg) add power management: - Dynamic gating based on busy signal - Configurable idle threshold - 0-cycle ungating latency - Best for packet-based streams with idle gaps

---

6.7 Performance Characteristics

6.7.1 Throughput

Configuration	Bandwidth	Notes
32-bit @ 400 MHz	1.6 GB/s	Typical DSP
64-bit @ 400 MHz	3.2 GB/s	Video processing
128-bit @ 400 MHz	6.4 GB/s	High-bandwidth video
512-bit @ 400 MHz	25.6 GB/s	Network/storage

6.7.2 Latency

Path	Cycles	Notes
Buffer latency	1	Skid buffer overhead
Master → Slave	2-3	Through interconnect
Typical pipeline	5-10	Multi-stage processing

6.7.3 Resource Usage

Module Type	LUTs	FFs	BRAM	Notes
Master/ Slave (64-bit)	~150	~100	0	Base module
Clock-gated (+cg)	+50	+30	0	Clock gating logic
Wider data (512-bit)	~400	~350	0	8x data width

vs AXI4: ~60-70% resource savings (no address channels, simpler protocol)

6.8 Common Use Cases

6.8.1 Video Processing

Characteristics: - High throughput (1-10 GB/s) - Regular packet structure (lines/frames) - TLAST marks line/frame boundaries - TUSER carries frame metadata

Recommended Configuration:

```
.AXIS_DATA_WIDTH(64-128),  // Multiple pixels per cycle
.SKID_DEPTH(4-5),          // 16-32 entry buffer
.AXIS_USER_WIDTH(8-16)     // Frame/line metadata
```

6.8.2 Network Packet Processing

Characteristics: - Variable packet sizes - High packet rate - TID for flow identification - TDEST for output routing

Recommended Configuration:

```
.AXIS_DATA_WIDTH(256-512), // Wide data bus
.SKID_DEPTH(5-6),          // 32-64 entry buffer
.AXIS_ID_WIDTH(8),         // 256 flows
.AXIS_DEST_WIDTH(6)        // 64 ports
```

6.8.3 DSP Pipelines

Characteristics: - Continuous data flow - Fixed sample rate - Minimal sideband signals - Low latency requirements

Recommended Configuration:

```
.AXIS_DATA_WIDTH(32-128),  // Sample data
.SKID_DEPTH(2-3),          // Minimal buffer
.AXIS_ID_WIDTH(0),         // No multiplexing
.AXIS_DEST_WIDTH(0),       // No routing
.AXIS_USER_WIDTH(4)        // Minimal metadata
```

6.9 Parameter Selection Guidelines

6.9.1 Data Width (AXIS_DATA_WIDTH)

Application	Recommended Width	Rationale
Audio processing	32-64 bits	1-2 samples per cycle

Application	Recommended Width	Rationale
SD video	64-96 bits	2-4 pixels per cycle
HD video	128-256 bits	Multiple pixels per cycle
4K video	256-512 bits	High pixel throughput
Network (1G)	64-128 bits	Moderate packet rate
Network (10G/100G)	256-512 bits	High packet rate

6.9.2 Buffer Depth (SKID_DEPTH)

SKID_DEPTH	Buffer Size	Use Case
2	4 entries	Low latency, continuous streams
3	8 entries	Typical DSP pipelines
4	16 entries	Video processing (default)
5	32 entries	Network packets, variable latency
6	64 entries	High-latency tolerance

6.9.3 Sideband Signal Widths

TID (Stream ID): - 0: Single stream, no multiplexing - 4: Up to 16 concurrent streams - 8: Up to 256 concurrent streams

TDEST (Destination): - 0: No routing (point-to-point) - 4: Up to 16 destinations - 6: Up to 64 destinations (network)

TUSER (User Metadata): - 0-1: Minimal or no metadata - 4-8: Frame/packet control - 16+: Complex metadata

---

6.10 Related Documentation

6.10.1 Protocol Specifications

- ARM IHI 0051A: AMBA AXI4-Stream Protocol Specification

6.10.2 RTL Design Sherpa Documentation

- **RTLAmba Overview** - Complete AMBA subsystem architecture
- **AXI4 Modules** - Memory-mapped protocol (comparison)
- **AXIL4 Modules** - Simplified control interface
- **GAXI Modules** - Generic AXI utilities (buffers, FIFOs)

6.10.3 Source Code

- RTL: `rtl/amba/axis4/`
- Tests: `val/amba/test_axis*.py`
- Framework: `bin/TBClasses/components/axis4/`
- Shared Infrastructure: `rtl/amba/shared/` (gaxi_skid_buffer)

6.11 Design Notes

6.11.1 When to Use AXI4-Stream

✓ **Use AXI4-Stream For:** - Video/image processing pipelines - Network packet processing - DSP data flows - High-throughput unidirectional data - Streaming accelerators

✗ **Use AXI4 For:** - Memory-mapped register access - Random access patterns - Bidirectional data flows - Address-based routing

6.11.2 Buffer Depth Trade-offs

Shallow Buffers (SKID_DEPTH = 2-3): - ✓ Lower latency - ✓ Smaller area - ✗ Less tolerance for backpressure - **Use for:** Low-latency DSP, continuous streams

Deep Buffers (SKID_DEPTH = 5-6): - ✓ High backpressure tolerance - ✓ Better throughput under variable load - ✗ Higher latency - ✗ Larger area - **Use for:** Network packets, variable-latency paths

6.11.3 Sideband Signal Optimization

Minimize unused signals for area/power:

```
// Video processing - no routing needed
.AXIS_ID_WIDTH(0),     // No stream multiplexing
```

```
.AXIS_DEST_WIDTH(0),   // No destination routing
.AXIS_USER_WIDTH(8)    // Only frame metadata

// vs Network processing - full routing
.AXIS_ID_WIDTH(8),     // 256 flows
.AXIS_DEST_WIDTH(6),   // 64 ports
.AXIS_USER_WIDTH(16)   // Full packet metadata
```

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6.12 Navigation

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